

From Reviewable to Reviewed

A Dialectical Standard of Care for Nurse- and Physician-Facing Clinical AI

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The Oversight Gap

Law already makes the clinician load-bearing

Clinical AI is no longer an unregulated frontier.

- The FDA lists **well over 1,000** AI-enabled devices [51].
- Of **1,016** analysed: **96.5%** cleared via **510(k)**; **<20%** cite prospective validation [30].
- EU: AI that is, or is a safety component of, a medical device needing notified-body assessment is **high-risk**; its oversight duties apply from **2 August 2028** [12, 13].

Neither regime asks for a human on every output

Each instead makes the **reviewing clinician** load-bearing at one defined point:

- **US**: exclusion from device rules
- **EU**: design-level oversight duties

Capability, not conduct

US: 21st Century Cures Act, CDS exclusion

Software escapes the device definition only where it enables the clinician “to independently review the basis” for its recommendations [47].

EU AI Act, Art. 14(4)(b) & Art. 26

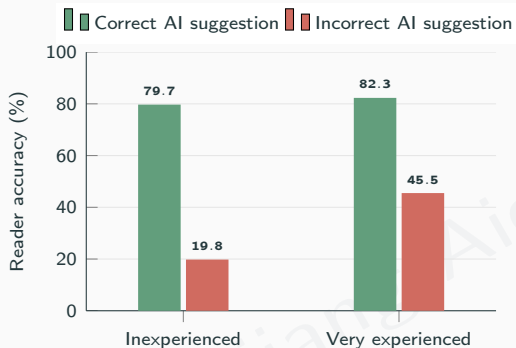
Overseers must be enabled “to remain aware” of **automation bias**; deployers keep system logs for ≥ 6 months [12].

What no provision requires

- ✗ judgment formed *independently* of the output
- ✗ that judgment *recorded*
- ✗ *consequences* when it is not

 A deployment can comply while every output is accepted unexamined.

The failure mode is documented



27 radiologists, mammography, purported AI BI-RADS suggestions [8]

- 265 clinicians: an “AI” label lowered experts’ quality ratings but changed no one’s behaviour [15].
- 220 clinicians: incorrect advice *with* feature-level explanations ✗ reduced accuracy [23].

Honest scope

Simulated cases, manipulated advice: they show the failure *can* be induced, not how often it occurs in deployment — a compliant deployment need not record whether review happened.

Our proposal: dialectical review

Definition

Dialectical review: an independently formed human assessment and a machine output are brought into **explicit confrontation**, and the resolution is recorded as a first-class clinical artifact.

Builds on

- Human-in-the-loop mandates rarely say *why* the human is there [6]; oversight can give false assurance [18].
- Cognitive forcing cuts overreliance, at a cost in acceptance [4].

Adds

- ★ An **evidentiary artifact**: the recorded disposition.
- ★ A **consequence rule** tying it to revalidation and to responsibility allocation.
- ★ The **clinician–AI dyad** as the unit bearing the standard of care.

Scoping the Argument

Three classes, three failure modes

Class	Representative deployment	Typical pathway	Characteristic failure	Oversight posture
A. Discriminative detection	Mammography CAD; autonomous retinopathy screening	510(k) ; De Novo (DEN180001) for autonomous	Miscalibrated confidence; distribution shift	Assistive: dialectical review. Autonomous: <i>excluded</i>
B. Risk prediction on EHR data	Proprietary sepsis prediction; early warning scores	Often non-device CDS or institutional tool	Calibration drift across sites; alert burden	Time-critical: D1 relaxed to a <i>post hoc</i> record
C. Generative & agentic	Ambient documentation; summarization; conversational medication management	Some device-regulated (e.g. 510(k) with PCCP); many not	Fabrication & omission; sycophancy to false premises	Dialectical review with provenance marking

💡 Shared invariant: expressed confidence is not a reliable guide to correctness.

Sources: [1, 55, 48, 50]

Three clarifications

1 “Hallucination”

Reserved for **Class C** [24]. A detector that outputs a BI-RADS category generates no text, so cannot fabricate a citation.

2 Probabilities

A softmax is a valid distribution, yet networks can be **miscalibrated** [20]. The interface usually shows a flag or a band — uncertainty is discarded.

3 Stochasticity

Temperature zero $\neq$ reproducible: batch-sensitive inference depends on server load [22]. A property of the **serving stack**.

→ The invariant across classes is not fabrication but **unwarranted expressed confidence**.

An operational definition of the dyad

The dyad

$$\langle h, S \rangle, \quad S = \langle m, v, c \rangle$$

h : a named, licensed clinician acting within scope m : model family v : released version c : institutional configuration (thresholds, interface, routing, escalation, population)

1. Same model, two hospitals, two thresholds → **two dyads**; evidence does not transfer without justification.
2. v can change under a **PCCP** [49]: the declared **version envelope** is a governance artifact.
3. The dyad performs the *act*; the **vendor fixes m, v and the institution fixes c** — neither's duty is discharged by it.

Two modes of reasoning, bounded

Machine side

- Meta-analysis, **83** studies: accuracy **52.1%**; no significant difference vs physicians overall ($p = \mathbf{0.10}$), **15.8** pp below experts [46].
- **2,400** real cases: LLMs worse than clinicians [21].
- RCT, **50** physicians: no gain ($p = \mathbf{0.60}$) [16].

Human side

Polanyi and Dreyfus [40, 9] do not make clinical judgment unformalizable: much of it is scored (BI-RADS, early warning scores). The residue is **indexical**: does this rule fit *this* patient, *now*?

Often asymmetric

If only one party knows the output is probabilistic or locally tuned, the conflict is not a dialectic but a **substitution**.

- 💡 The proposal is **procedural, not psychological**: we do not rely on debiasing clinicians (whether biases drive error is itself contested [36]).

Architecture as Regulation

Two claims about architecture, usually run together

Lessig: *regulative effect*

Architecture regulates conduct ex ante and self-executes [29].

Judge et al.: *auditability*

Trained networks are “opaque and not designed”; compliance with a specification cannot be demonstrated [25].

An unauditable model can still **constrain its user**.

What survives: the **deployment layer**

- which question is put to the model
- what threshold fires an alert
- ranked differential vs. single answer
- uncertainty shown or discarded
- where the alert routes
- whether an override demands a reason

💡 Consequential *and* specifiable — regulation should attach here.

Two regimes, one object

Device regulation and professional negligence law now share an object, with **bidirectional scope dependence**:

- CDS eligibility turns on what the clinician *can do* with the basis [47].
- Whether an authorized tool shapes the **standard of care** is a question malpractice law has so far resisted [42].

Vendor: arbitrage

Claim non-device CDS (“clinicians review the basis”), then market on validated performance — a proposition about **reliance**.

Clinician: chilling

Liability pushes AI toward “confirmatory advice only”, which “leaves unrealised some of the value of AI” [43].

Structurally akin to Solow-Niederman’s *inter-regime doctrinal collapse* [45]: unmanaged, not inherently bad.

Confident error: a bounded analogy

The claim is **functional, not mechanistic**: both errors are confidently expressed and invisible at the point of use.

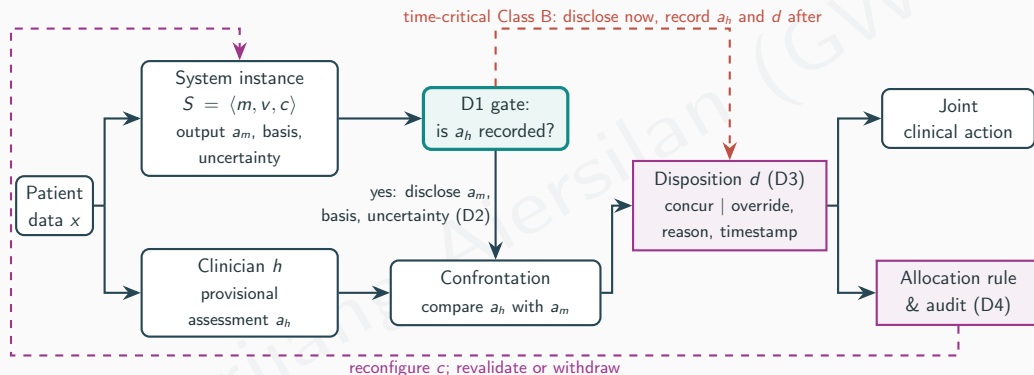
Failure mode	Human form	Machine analogue
Anchoring	Early framing fixes the differential [7]	Output varies with presentation order [21]
Confirmation bias	Consistent findings weighted more heavily	Sycophancy toward false premises [38]
Premature closure	Leading cognitive contributor in 100 error cases [17]	Single recommendation, no differential [52]

→ Shared **incentive architecture** penalizes expressed uncertainty — a common point of intervention, not a common cause [26].

⚠ Retrieval helps but does not eliminate it [54]: **1.47%** hallucination, **3.45%** omission over **12,999** sentences; **44%** of hallucinations could affect care [3].

Dialectical Review

Dialectical review as a specified procedure



Recording a_h before a_m is shown means sequence, not exhortation, controls anchoring.

D1 and D2: an independent prior, a usable basis

D1 Independent prior

The clinician records a_h *before* a_m is disclosed.

- Anchoring works through sequence, so **sequence is the control variable** — unlike Art. 14(4)(b)'s mere awareness.
-  Class B: relaxed to a *post hoc* record. Not independent, possibly anchored; audit should weight it so.

D2 Basis at decision-relevant granularity

What a clinician needs *in order to disagree*:

- **System level**: population, operating point, calibration, known failure conditions.
- **Case level**: inputs used, missing data, case-specific uncertainty; for Class C, sources and tools.
- Rationale *last*: it hurt accuracy when advice was wrong [23].

D3 and D4: a record, and a consequence

D3 Recorded disposition

A structured record d : **concur or override**, a reason code, a timestamp.
The only condition that records how the confrontation was resolved.

Agentic systems: the unit is the **committed action** — anything that writes to the record, places an order, or reaches a patient.

D4 Consequence rule (monitoring half)

Per system instance and interval, stratified by unit and patient group:

- **override rate** — share of dispositions disagreeing with a_m
- **pro forma share** — concurrences whose reason is the modal default

Bands are preregistered before go-live [34]. Crossing one, either way, triggers: reassess → reconfigure c → suspend or withdraw.

 Without the trigger, recording is merely archival.

The reflexivity objection

Dialectical review is itself architecture, so it is not immune to the deference it targets. **We accept this and narrow the claim.**

Measured costs

- **19** veterinary radiologists: responding first cut agreement with AI **whether or not it was right**; those who disagreed sought a colleague's second opinion less [14].
- Cognitive forcing cuts overreliance but is rated least favourably where it works best (lay tasks) [4].

What we do claim

- **Observability**: gamed logs leave traces — near-unanimous concurrence, sub-second dispositions, one dominant reason code.
- **Falsifiability**: it fails if D1–D4 deployments, with honest logs, do no better than matched deployments without them.

No such study exists yet; we do not claim its result.

Allocating responsibility

The allocation rule states duties; the D3 record is the **evidence** of their discharge. Duty, breach, and causation stay with doctrine.

Vendor

Validated performance within intended use and change envelope; disclosure enabling D2. The EU **Product Liability Directive** covers software; transposition by **9 Dec 2026** [11]. US status unsettled [31, 43].

Institution

Local validation; configuration of c; alert burden; retain outputs for every monitored patient, linked to outcomes; staffing and competency [32, 41].

Clinician

Perform the review; record an honest disposition. A reasoned override that proves wrong is **ordinary clinical judgment**. A missing disposition is a breach.

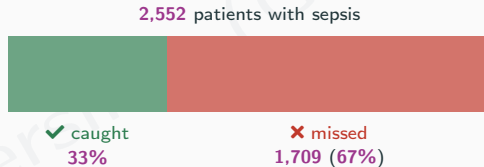
→ Hard cases: a pro forma concurrence earns **✗ no defensive weight**; override rate is **✗ never an individual metric**; good-faith dispositions need a **✓ safe harbor**. EU AI Liability Directive withdrawn Oct 2025 [10].

Nursing: Where the Dyad Operates

Worked case: proprietary sepsis alerting

External validation [55]: **27,697** patients,
38,455 hospitalizations.

- AUROC **0.63** (95% CI **0.62–0.64**)
- Threshold **6** (from vendor's range):
sensitivity **33%**, PPV **12%**
- Alerts in **18%** of hospitalizations
- **≈8** patients evaluated per case found



Why it matters

Sepsis is one of five conditions behind **38.7%** of serious diagnostic harms; an estimated **795,000** Americans die or are permanently disabled each year [35].

The hardest case for the framework

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- D1 Post hoc only — but nurses' disagreement becomes **data**: the patient who is visibly unwell while the score reads zero [33].
 - D2 **✗ Foreclosed by proprietary design**: the validation population was unavailable; the deficit surfaced only years later. A **vendor disclosure failure**.
 - D3 Makes alert burden and disagreement examinable — but **no log can see the 67% never alerted**. That needs scores retained for every patient, linked to outcomes.
 - D4 Puts that computation on a **preregistered schedule**, forces revalidation, and gives the nurse who escalates against a reassuring score a recorded defense.
-

⚠ Weak D3 already exists: medication alerts are overridden **49%–96%** of the time, governing nothing [53].

➔ Alert burden turns **nominal oversight into null oversight** [27, 44].

Competency: an act can be taught

D1–D3 are **observable behaviours**, so they can be entrusted, supervised, and evaluated.

Nurses already disagree with scores

305 registered nurses [28]:

- **40%** felt confident overriding the escalation protocol
- only **3.6%** said early warning tools do not impede higher-order thinking

Infrastructure is only partly ready

- **AACN Essentials** (2026): AI still not named in the informatics sub-competencies [2].
- Delphi, **48** experts: **10** domains, **61** sub-competencies for AI-driven monitoring in long-term care [19].

💡 Missing: a specification of the **reviewing act** to organize them around.

A distributive precondition

Global nursing workforce, 2023 [56]:

- **29.8 million** nurses
- shortage of **5.8 million**, projected **4.1 million** by **2030**
- **78%** of nurses work in countries holding **49%** of the world's population

Otherwise, an unfunded mandate

Where the shortage is deepest, pressure to substitute is greatest and slack to review is least.

So: an institutional duty first

Configuring *c* to produce an alert burden staff cannot review is not delegation — it is a foreseeable failure.

Patients and Limits

The patient: autonomy, and a limit on justice

Autonomy

US informed-consent doctrine generally attaches no liability to not telling patients that ML informed a recommendation [5]. Yet of **11,004** US adults, **60%** would be uncomfortable with a provider relying on AI, and **57%** expect it to worsen the relationship [39].

→ D3 is a precondition of **substantive disclosure**.

Justice: what the dyad cannot do

By industry estimates, such algorithms score some **200 million** Americans a year. One gave equal risk to Black and White patients who were not equally sick; correcting it would lift Black patients flagged for extra help from **17.7%** to **46.5%** [37].

⚠ Each case looked plausible:
bedside review cannot see it.

Stratified disposition rates give D4 an **aggregate** signal — but the primary duty is population-level.

Limits, stated plainly

What the proposal offers

- ✓ Makes review observable
- ✓ Clarifies who owes what
- ✓ Gives disclosure real content

What it costs or cannot do

- ✗ Certain cost in clinician time
- ✗ D1 cuts appropriate reliance too [14]
- ✗ Assistive only; US/EU law only
- ✗ Reason codes can collapse into defaults

💡 The warranted conclusion: **piloted policy under evaluation**, not a general mandate.
An assistive system with near-unanimous concurrence may be **functionally autonomous**.

Conclusion

Takeaways

1. ★ **Reviewable** $\neq$ **reviewed** — both regimes can be met with no review at all.
2. ★ **Dialectical review** — D1 independent prior, D2 basis, D3 recorded disposition, D4 consequence rule.
3. ★ **The dyad** $\langle h, S \rangle$ — bears the standard of care; the record allocates responsibility.

⚠ Auditable oversight is not *effective* oversight

⚠ Bedside review cannot detect population-level bias

⚠ Review costs time: an institutional duty first

In one sentence

A checkable specification of what **being in the loop** must mean.

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Thank you!

Questions?

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Backup: the evidence base at a glance

Study	Sample	Finding used in the talk
Dratsch et al. [8]	27 radiologists	Accuracy 82.3% → 45.5% (experienced) under incorrect suggestions
Gaube et al. [15]	265 clinicians	"AI" label changed perceived quality, not behaviour
Jacobs et al. [23]	220 clinicians	Wrong advice + explanations reduced accuracy
Fogliato et al. [14]	19 vet. radiologists	Responding first cut agreement regardless of accuracy
Wong et al. [55]	27,697 patients	Sepsis model AUROC 0.63; 67% of cases missed
Le Lagadec et al. [28]	305 nurses	40% confident to override protocol
Pew Research Center [39]	11,004 US adults	60% uncomfortable with provider reliance on AI
Obermeyer et al. [37]	~200M/yr scored	17.7% → 46.5% after correcting target bias